

Solution Requirements

Date	25 June 2026
Team ID	
Project Name	Rising Waters – A Flood Prediction
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow registered users to securely log in using their email and password before accessing flood prediction services.	High
2	Authorization levels	The system shall provide role-based access where administrators manage the ML model and users perform flood risk predictions.	High
3	External interfaces	The system shall integrate with the Kaggle flood dataset, Flask web framework, Joblib/Pickle model files, and IBM Cloud deployment services.	Medium

4	Transactions processing	The system shall validate user inputs, preprocess weather parameters, execute the trained ML model, and return flood prediction results instantly.	High
5	Reporting	The system shall display prediction results, maintain prediction history, and allow users to view or download prediction reports.	Medium
6	Business rules, etc.	Flood prediction shall only be generated after validating all required weather parameters. The trained XGBoost model shall be used for prediction.	High
7	Compliance to laws or regulations	The system shall securely handle user information following standard data privacy and cybersecurity practices.	Medium
8	Other	The system shall support future ML model updates and additional weather parameters without affecting existing functionality.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application shall process user inputs and generate flood prediction results quickly.	Prediction generated in less than 3 seconds.
2	Scalability	The application shall efficiently support increasing numbers of users and larger environmental datasets.	Supports 500+ concurrent users without performance degradation.
3	Security & Data Privacy	User credentials and prediction data shall be securely stored and transmitted.	Password encryption, secure authentication, and HTTPS communication.
4	Reliability & Availability	The system shall consistently provide accurate flood predictions with minimal downtime.	99% uptime during normal operation.
5	Usability & Accessibility	The application shall provide a simple, intuitive, and responsive interface for all users.	Users can complete a prediction within 2 minutes on desktop or mobile devices.
6	Other	The application shall be modular, easy to maintain, and deployable across multiple platforms.	Deployable on Windows, Linux, macOS, and IBM Cloud with easy ML model updates.